Trash Detection and Environment Cleanliness Analysis using YOLOv8 and an Interactive Assistant

Course Project Report

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Project Focus: Detect litter, visualize trash hotspots, and estimate cleanliness.

Abstract

This project presents a deep-learning based trash detection and cleanliness analysis system. The main goal was not only to detect litter objects in real-world images, but also to make the output useful and understandable for a normal user. For this purpose, YOLOv8 object detection was used to identify different trash categories and draw bounding boxes around them. After detection, the system generates a heatmap overlay to show where litter is concentrated and calculates a cleanliness score from 0 to 100. A Gradio-based interactive assistant was also developed so that users can upload an image and receive the detection output, heatmap, cleanliness level, and a short explanation.

The dataset used in this project was a TACO-style litter detection dataset converted into YOLO format. It contained 59 trash categories including bottles, cans, wrappers, cups, plastic bags, plastic film, cigarette waste, and unlabeled litter. The final dataset split contained 1200 training images and 299 validation images. Multiple YOLOv8 models were tested, starting from YOLOv8n as a baseline and later improving the pipeline using YOLOv8s and YOLOv8m. The final model used YOLOv8m with higher image resolution, stronger augmentation, AdamW optimizer, and saved best checkpoint evaluation.

Overall, the project produced a complete pipeline: dataset preparation, model training, validation, prediction visualization, heatmap generation, cleanliness scoring, and an interactive demo. The result is a practical AI assistant that can support environmental monitoring by detecting visible litter and giving an easy-to-understand cleanliness interpretation.

1. Introduction

Trash and litter in public areas are common environmental problems. Manual monitoring is slow, inconsistent, and difficult to scale because it depends on people physically checking streets, parks, beaches, sidewalks, and campus areas. A computer vision system can help by automatically detecting visible litter from images and giving a quick summary of how polluted an area looks.

The main aim of this project is to build a practical AI system that can detect trash objects in real-world images using deep learning. Instead of only showing bounding boxes, the project also adds a heatmap and cleanliness score so that the result is easier to understand. This makes the system more useful for non-technical users because they can immediately see where the trash is located and how severe the scene appears.

The final system performs four main tasks:

- Detects trash categories using YOLOv8 object detection.
- Localizes litter using bounding boxes.
- Highlights polluted regions using a spatial heatmap overlay.

- Summarizes the scene using a 0–100 cleanliness score and assistant-style explanation.

2. Background

2.1. Object Detection

Object detection is different from simple image classification. In classification, the model only predicts what is present in the image. In object detection, the model predicts both the object class and the location of the object. This location is usually represented using a bounding box.

For example, if an image contains a plastic bottle and a food container, a classification model may only say that the image contains trash. An object detection model can show exactly where the plastic bottle and food container are located. This is important for trash monitoring because the location and spread of litter matter.

2.2. YOLOv8

YOLO stands for "You Only Look Once." It is a one-stage object detection method that performs localization and classification in one forward pass. YOLO models are commonly used because they are fast and suitable for real-time or near-real-time applications.

In this project, YOLOv8 was selected because it is easy to train, supports custom datasets, provides useful evaluation plots, and works well with bounding-box based detection problems. The final project used YOLOv8m because it provided more model capacity for the 59-class litter detection task.

3. Dataset Description

The project used a TACO-style litter detection dataset prepared in YOLO format. The dataset contained realistic outdoor scenes such as roads, grass, sand, leaves, sidewalks, and public spaces. This made the task challenging because litter objects can be small, partially hidden, or visually similar to the background.

The prepared dataset contained:

- 59 trash classes.
- 1200 training images.
- 299 validation images.
- YOLO label files containing class IDs and normalized bounding box coordinates.

Examples of classes include plastic bottles, clear plastic bottles, drink cans, food containers, plastic bags, plastic film, cigarette waste, cups, wrappers, and unlabeled litter.

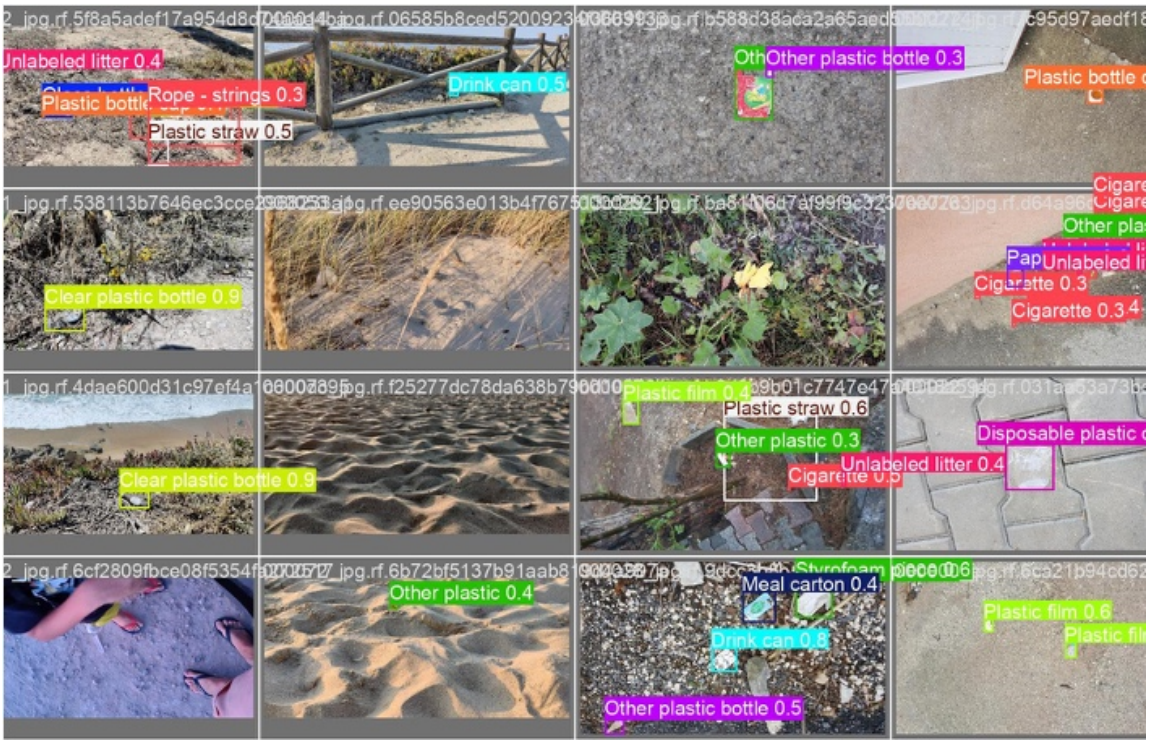


Figure 1: Sample validation predictions showing multiple litter categories such as plastic bottles, drink cans, cigarettes, wrappers, and unlabeled litter. This screenshot was used in the presentation and is included here as part of the report evidence.

4. Methodology

The project followed a complete deep-learning workflow from dataset preparation to user-facing deployment.

4.1. Dataset Preparation

First, the dataset was mounted from Google Drive and copied into the Colab working environment. The dataset folders were inspected to verify that the training and validation images existed. A fixed YOLO data YAML file was created or verified so that the model could correctly find the training and validation folders.

The code checked the number of images in each split before training. This step was important because incorrect paths or missing labels can cause invalid training results. After verification, the dataset was ready for YOLOv8 training.

4.2. Model Training

The project did not rely on only one model. Three YOLOv8 configurations were explored:

- **YOLOv8n Baseline:** A lightweight model used for the first experiment.
- **YOLOv8s Improved:** A stronger model with higher resolution and augmentation.

- **YOLOv8m Final:** The final model with higher capacity for the 59-class detection problem.

The final YOLOv8m model was trained using the following important settings:

- Pretrained checkpoint: `yolov8m.pt`
- Image size: 832
- Batch size: 4
- Optimizer: AdamW
- Initial learning rate: 0.0006
- Cosine learning-rate schedule
- Weight decay: 0.0005
- Strong augmentation including HSV changes, translation, scaling, shear, flipping, mosaic, and mixup

These settings were selected because trash objects are often small, cluttered, and affected by lighting and background variation. Higher image size helped the model see small objects more clearly, while augmentation improved robustness.

4.3. Evaluation

After training, the best checkpoint was loaded and evaluated on the validation split. The project generated training curves, validation curves, prediction images, and confusion matrices. These outputs helped us understand whether the model was learning and where it was still confused.

5. Model Development and Comparison

The model improved step by step from the baseline to the final version. The baseline YOLOv8n model was useful for checking whether the dataset and training setup worked. The improved YOLOv8s model used higher image resolution and augmentation. The final YOLOv8m model increased model capacity and used the saved best checkpoint for final evaluation.

Table 1: Comparison of YOLOv8 models used in the project.

Model	Epochs	Image Size	Precision	Recall	mAP@50	Notes
YOLOv8n Baseline	30	640	0.480	0.170	0.140	Baseline model
YOLOv8s Improved	57	960	0.394	0.224	0.181	Higher resolution and augmentation
YOLOv8m Final	86	832	0.413	0.194	0.183	Stronger model and best checkpoint

The final model did not only focus on one metric. Precision, recall, mAP, qualitative detections, and assistant usability were considered together. The results show that the model learned useful patterns, but the task remained challenging due to many visually similar classes and small objects.

6. Results and Visual Evidence

6.1. Training Curves

The training results show that the model learned from the dataset. The box loss, classification loss, and DFL loss generally decreased as training progressed. This means the model became better at locating objects, predicting classes, and refining bounding boxes.

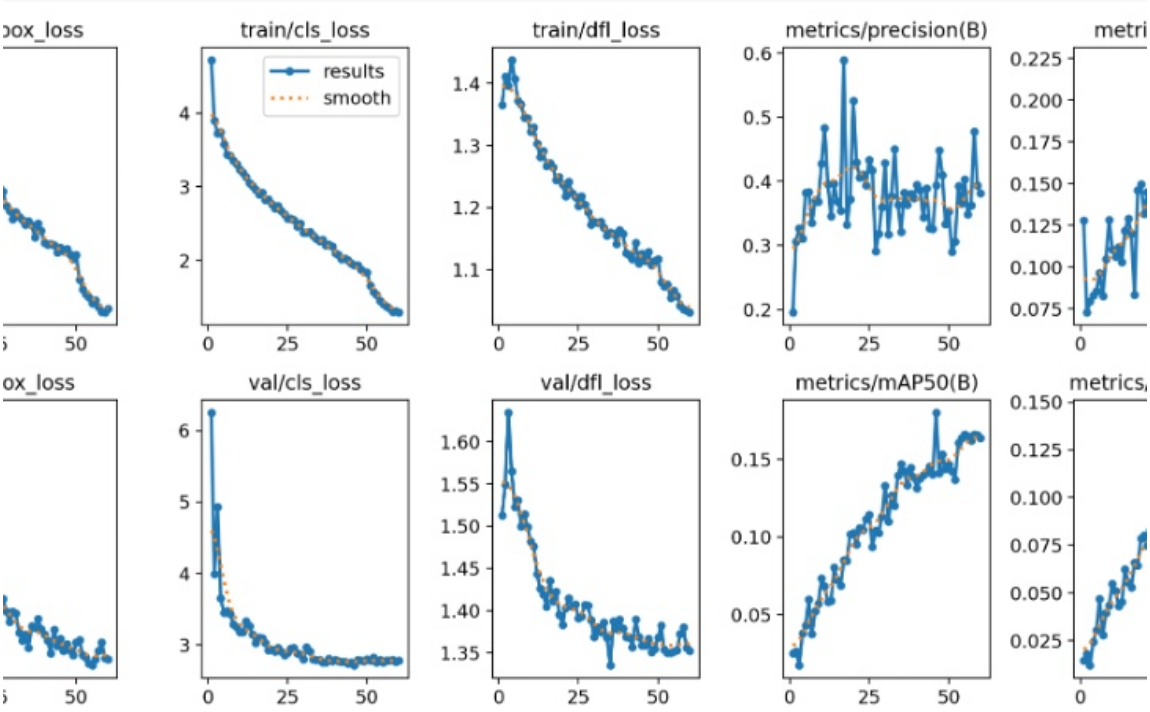


Figure 2: Training and validation curves from the YOLOv8 experiment. Losses decrease over time, while precision, recall, and mAP trends show model improvement during training.

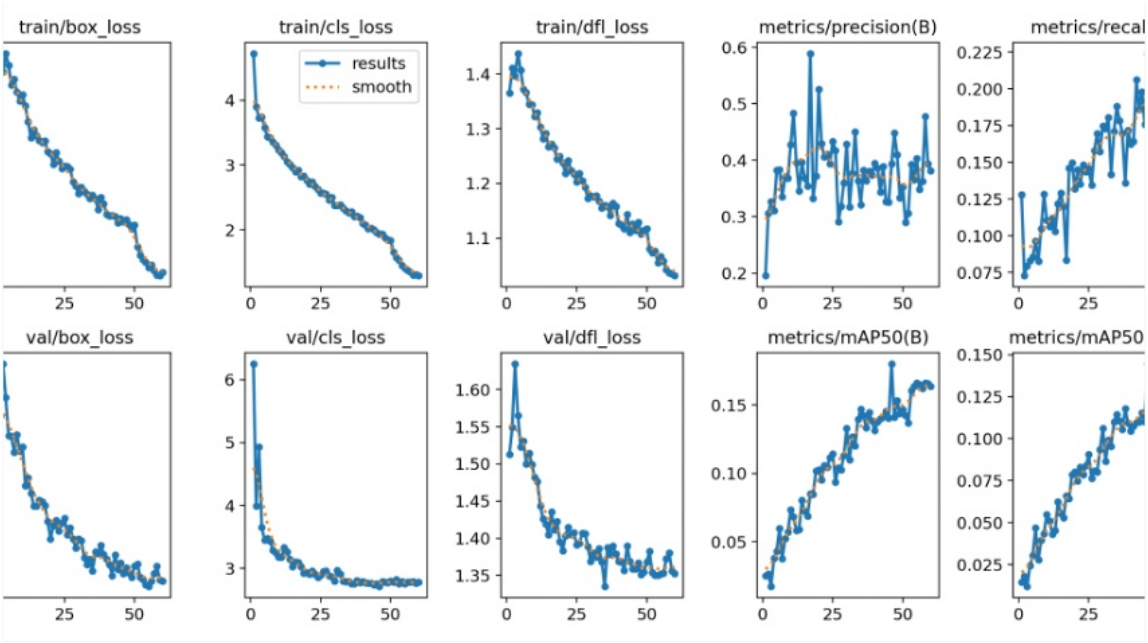


Figure 3: Final training results plot included in the presentation. The decreasing loss curves indicate that the model was learning meaningful features from the litter dataset.

6.2. Single Image Detection Example

A single validation image was tested using the final YOLOv8m checkpoint. The model detected two trash-related objects: a single-use carrier bag and food waste. The output shows that the model can localize visible objects using bounding boxes.



Figure 4: Final YOLOv8m trash detection example. The model detects a single-use carrier bag and food waste in the uploaded validation image.

6.3. Detection, Heatmap, and Cleanliness Output

The project added extra interpretation after object detection. The heatmap highlights the spatial area where trash was detected. The cleanliness score was calculated using object count, object size, and confidence. In this example, the image received a score of 0/100 and was labeled as highly polluted.

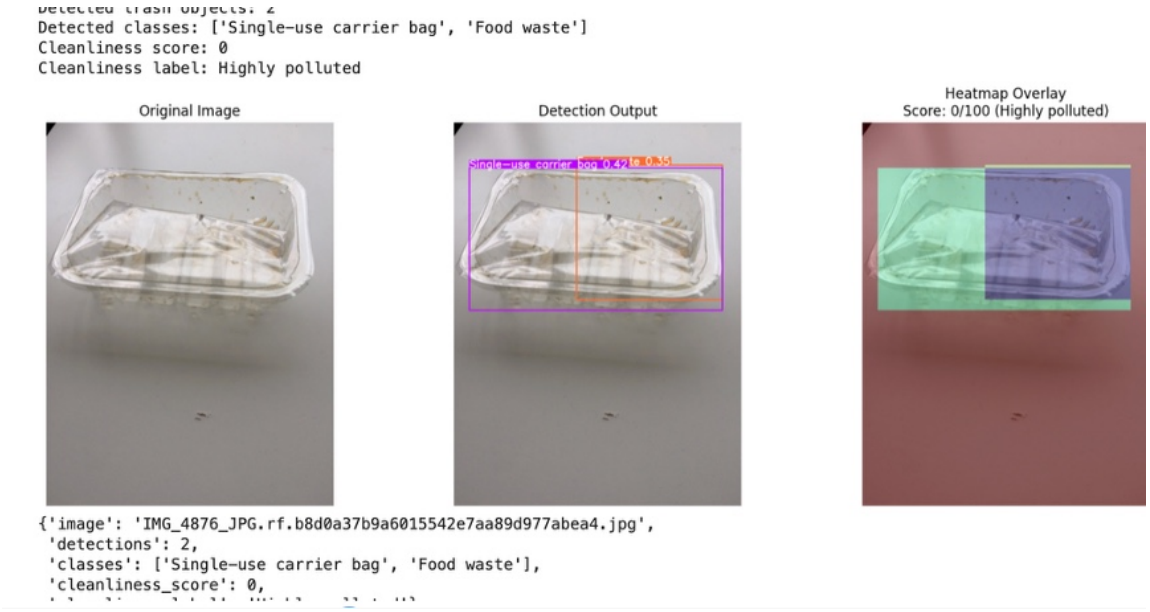


Figure 5: Original image, detection output, and heatmap overlay. The assistant detected two objects and assigned the scene a cleanliness score of 0/100, labeled as highly polluted.

6.4. Confusion Matrix

The confusion matrix helped diagnose which classes were being predicted correctly and which classes were confused. Since the dataset contains many similar trash categories, class-level confusion is expected. For example, different plastic categories, wrappers, and food containers can look similar in real-world scenes.

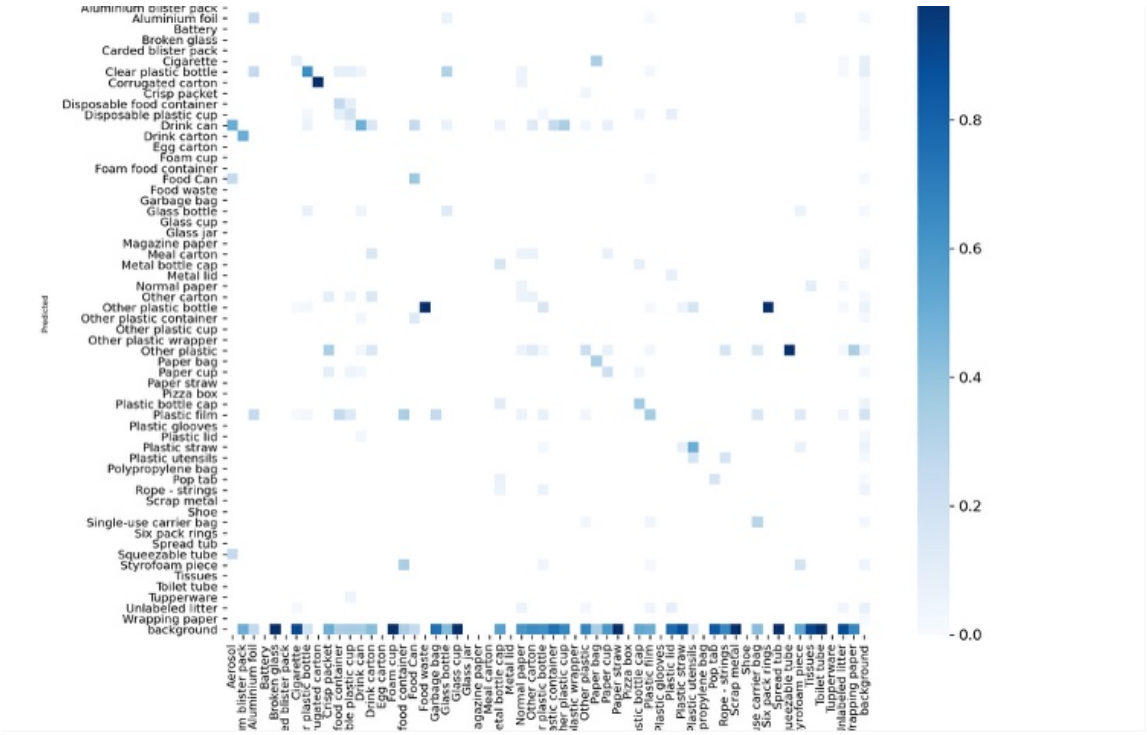


Figure 6: Confusion matrix for the litter detection model. It shows class-level prediction behavior and highlights confusion among visually similar trash categories.

7. Value Additions

A normal YOLO project usually stops after training and bounding-box prediction. This project added three important value additions.

7.1. Heatmap Overlay

The heatmap converts detection boxes into a visual pollution hotspot map. Each detected box contributes to the heatmap based on its confidence. Areas with more detected trash appear as stronger polluted regions. This makes it easier to understand where litter is concentrated.

7.2. Cleanliness Score

The cleanliness score gives a simple 0–100 interpretation of the image. A higher score means the scene appears cleaner, while a lower score means the scene appears more polluted. The score uses three ideas:

- number of detected objects,
- size of the detected objects compared with the image area,
- confidence of the detections.

The scoring idea can be summarized as:

$$\text{Cleanliness} = 100 - \text{Pollution Score}$$

where the pollution score increases when there are more objects, larger objects, and higher-confidence trash detections.

7.3. Interactive Assistant

The final assistant was built using Gradio. A user can upload an image and receive three outputs: detection image, heatmap overlay, and assistant summary. This makes the project more practical because the model is not limited to notebook-only testing.



Figure 7: Interactive Trash Detection and Cleanliness Assistant. The interface shows the uploaded image, detection output, heatmap overlay, and assistant-generated summary.

8. Assistant Pipeline

The final assistant works in the following order:

1. The user uploads an image.
2. The image is passed to the final YOLOv8m model.
3. The model predicts bounding boxes, classes, and confidence scores.

4. The detection output is drawn on the image.
5. The detected boxes are used to generate a heatmap overlay.
6. The cleanliness score and cleanliness label are calculated.
7. A short assistant summary is generated for the user.

The cleanliness labels used in the project were:

- **Clean:** score 80 or above.
- **Moderately littered:** score from 50 to 79.
- **Polluted:** score from 25 to 49.
- **Highly polluted:** score below 25.

9. Challenges

This project had several technical challenges.

9.1. Class Imbalance

Some trash categories appeared more often than others. This makes training harder because the model gets more examples of common classes and fewer examples of rare classes.

9.2. Small Object Detection

Many trash objects are small, far away, or partially covered. Small objects are harder to detect because they occupy fewer pixels in the image.

9.3. Similar-Looking Classes

Several categories look visually similar, such as plastic film, wrappers, plastic bags, and different bottle types. This can cause class confusion even when the model correctly finds the object location.

9.4. Cluttered Real-World Backgrounds

The dataset includes grass, sand, leaves, dirt, roads, and sidewalks. These backgrounds can hide trash objects and make detection more difficult.

9.5. Colab GPU and Runtime Limits

Training deep-learning models in Colab can be affected by GPU availability and runtime disconnections. To handle this, saved checkpoints were used so that the best model could still be evaluated and reported.

10. Future Work

The project can be improved in several ways:

- Collect more balanced data for rare classes.
- Use more small-object focused augmentation.
- Train with higher image resolution when more GPU resources are available.
- Improve the cleanliness scoring formula using real human ratings.
- Add GPS/location metadata for real environmental monitoring.
- Deploy the assistant as a web or mobile application.
- Add before-and-after comparison to measure cleanup progress.

11. Conclusion

This project built a complete trash detection and environmental cleanliness analysis system. The model was trained on a 59-class litter dataset using YOLOv8 and evaluated using validation metrics, prediction images, and confusion matrices. The final system goes beyond basic object detection by adding heatmap visualization, cleanliness scoring, and an interactive assistant.

The results show that YOLOv8 can detect and localize many trash objects in realistic scenes. At the same time, the project also shows the difficulty of the problem, especially because of small objects, class imbalance, and visually similar trash categories. The final assistant makes the output easier to understand by showing not only what trash was detected, but also where it is concentrated and how polluted the scene appears.

Overall, the project demonstrates a practical AI pipeline that can support environmental monitoring and can be extended into a real web or mobile tool in the future.